Math Class

The `Math` class in Java is part of the `java.lang` package and provides a variety of mathematical methods to perform basic and advanced mathematical operations. It includes methods for performing trigonometric calculations, logarithmic calculations, random number generation, and more.

Key Features of the Math Class

- All methods in the `Math` class are **static**, meaning you can call them directly from the class without creating an object of the `Math` class.
- It includes methods for **basic arithmetic operations**, **power and logarithmic functions**, **trigonometric functions**, **random number generation**, and **rounding**.

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1. Basic Math Operations

`abs()` - Absolute Value

Returns the absolute value of a number.

Syntax:

```
Math.abs(int a)
Math.abs(double a)
Math.abs(long a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        int x = -10;
        double y = -5.5;

        System.out.println("Absolute value of x: " + Math.abs(x));  // 10
        System.out.println("Absolute value of y: " + Math.abs(y));  // 5.5
    }
}
```

`max()` - Maximum of Two Numbers

Returns the larger of two numbers.

Syntax:

```
Math.max(int a, int b)
Math.max(double a, double b)
Math.max(long a, long b)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        int a = 10, b = 20;
        double x = 5.5, y = 7.5;

        System.out.println("Max of a and b: " + Math.max(a, b));  // 20
        System.out.println("Max of x and y: " + Math.max(x, y));  // 7.5
    }
}
```

`min()` - Minimum of Two Numbers

Returns the smaller of two numbers.

Syntax:

```
Math.min(int a, int b)
Math.min(double a, double b)
Math.min(long a, long b)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        int a = 10, b = 20;
        double x = 5.5, y = 7.5;

        System.out.println("Min of a and b: " + Math.min(a, b));  // 10
        System.out.println("Min of x and y: " + Math.min(x, y));  // 5.5
    }
}
```

`pow()` - Power

Returns the value of the first argument raised to the power of the second argument.

Syntax:

```
Math.pow(double base, double exponent)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double base = 2, exponent = 3;
        System.out.println("2 raised to the power 3: " + Math.pow(base, exponent));  // 8.0
    }
}
```

`sqrt()` - Square Root

Returns the square root of a number.

Syntax:

```
Math.sqrt(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double number = 16.0;
        System.out.println("Square root of 16: " + Math.sqrt(number));  // 4.0
    }
}
```

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2. Rounding and Truncating

`round()` - Round to the Nearest Integer

Rounds a floating-point number to the nearest integer.

Syntax:

```
Math.round(float a)
Math.round(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        float number1 = 5.6f;
        double number2 = 3.3;

        System.out.println("Round of 5.6: " + Math.round(number1));  // 6
        System.out.println("Round of 3.3: " + Math.round(number2));  // 3
    }
}
```

`ceil()` - Ceiling (Round Up)

Returns the smallest integer that is greater than or equal to the argument.

Syntax:

```
Math.ceil(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double number = 5.3;
        System.out.println("Ceiling of 5.3: " + Math.ceil(number));  // 6.0
    }
}
```

`floor()` - Floor (Round Down)

Returns the largest integer that is less than or equal to the argument.

Syntax:

```
Math.floor(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double number = 5.8;
        System.out.println("Floor of 5.8: " + Math.floor(number));  // 5.0
    }
}
```

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3. Trigonometric Methods

`sin()` - Sine

Returns the sine of a number in radians.

Syntax:

```
Math.sin(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double angle = Math.PI / 2;  // 90 degrees in radians
        System.out.println("Sine of 90 degrees: " + Math.sin(angle));  //
1.0
    }
}
```

`cos()` - Cosine

Returns the cosine of a number in radians.

Syntax:

```
Math.cos(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double angle = Math.PI;  // 180 degrees in radians
        System.out.println("Cosine of 180 degrees: " + Math.cos(angle));  //
-1.0
    }
}
```

`tan()` - Tangent

Returns the tangent of a number in radians.

Syntax:

```
Math.tan(double a)
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        double angle = Math.PI / 4;  // 45 degrees in radians
        System.out.println("Tangent of 45 degrees: " + Math.tan(angle));  //
1.0
    }
}
```

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4. Random Number Generation

`random()` - Generate Random Number

Generates a random number between 0.0 (inclusive) and 1.0 (exclusive).

Syntax:

```
Math.random()
```

Example:

```
public class MathExample {
    public static void main(String[] args) {
        // Generate a random number between 0 and 100
```

```java
        int randomNumber = (int) (Math.random() * 100);
        System.out.println("Random number: " + randomNumber);
    }
}
```
